

PROJECT 1: DENSITY ESTIMATION AND CLASSIFICATION

CSE 575: STATISTICAL MACHINE LEARNING

A PROJECT REPORT

Submitted by

AKSHAY DESHNENI (ASU ID: 1218717838)



ARIZONA STATE UNIVERSITY

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Results of Project

In this project an attempt has been made to predict the handwritten digits 0 and 1 from the MNIST dataset using a Naïve Bayes Classifier. The classifier has been implemented from scratch using the Python programming language and some of its common libraries like NumPy, SciPy and Pandas. The classifier was trained using the training data set which consisted of 5000 samples each for Digit 0 and Digit 1. The test set was then fed to the trained classifier which consisted of 980 samples for Digit 0 and 1135 samples for Digit 1. The following eight parameters and their values have been listed below along with their standard deviations.

Mean_Feature1_Digit0 -> 44.2425
Variance_Feature1_Digit0 -> 114.826265
StdDev_Feature1_Digit0 -> 10.715702
Mean_Feature2_Digit0 -> 87.531654
Variance_Feature2_Digit0 -> 100.982534
StdDev_Feature2_Digit0 -> 10.049007
Mean_Feature1_Digit1 -> 19.370859
Variance_Feature1_Digit1 -> 30.871623
StdDev_Feature1_Digit1 -> 5.556224
Mean_Feature2_Digit1 -> 61.422686
Variance_Feature2_Digit1 -> 81.980445
StdDev_Feature2_Digit1 -> 9.054305

The formula for the Naïve Bayes Classifier used for calculating $P(X_i | Y)$ is given below:

For Digit 0:

```
exponent1 = (np.exp(-np.power(Dataset.iloc[index, 0] - mean_feature1_digit0.iloc[0], 2.)  
              / (2 * np.power(std_feature1_digit0.iloc[0], 2.))))  
exponent2 = (np.exp(-np.power(Dataset.iloc[index, 1] - mean_feature2_digit0.iloc[0], 2.)  
              / (2 * np.power(std_feature2_digit0.iloc[0], 2.))))  
 $P(X_i | Y = 0) = ((\text{exponent1} / (\text{np.sqrt}(2 * \text{np.pi}) * \text{std\_feature1\_digit0.iloc}[0])))$   
 $\quad * ((\text{exponent2} / (\text{np.sqrt}(2 * \text{np.pi}) * \text{std\_feature2\_digit0.iloc}[0])))$ 
```

For Digit 1:

```
exponent1 = (np.exp(-np.power(Dataset.iloc[index, 0] - mean_feature1_digit1.iloc[0], 2.)  
              / (2 * np.power(std_feature1_digit1.iloc[0], 2.))))  
exponent2 = (np.exp(-np.power(Dataset.iloc[index, 1] - mean_feature2_digit1.iloc[0], 2.)  
              / (2 * np.power(std_feature2_digit1.iloc[0], 2.))))  
 $P(X_i | Y = 1) = ((\text{exponent1} / (\text{np.sqrt}(2 * \text{np.pi}) * \text{std\_feature1\_digit1.iloc}[0])))$   
 $\quad * ((\text{exponent2} / (\text{np.sqrt}(2 * \text{np.pi}) * \text{std\_feature2\_digit1.iloc}[0])))$ 
```

Prior Probability for $P(Y = 0)$ and $P(Y = 1)$ was assumed to be 0.5.

Predicted Y for digit 0 $P(y0hat) = P(Y = 0) * P(X_i | Y = 0)$

Predicted Y for digit 1 $P(y1hat) = P(Y = 1) * P(X_i | Y = 1)$

Accuracy of Test Set for Digit 0 -> 0.9173469387755102 (approximately 92 percent)

Accuracy of Test Set for Digit 1 -> 0.9233480176211454 (approximately 92 percent)